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# Search-Guided Evolutionary Optimization for Graph Pathfinding

A comparative study of classical and hybrid search methods

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**Project type:** Reproducible course project

## Abstract

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This project studies route planning on deterministic obstacle grids through a common experimental interface. Six classical methods, including uninformed, cost-based, and heuristic search, are compared with a Genetic Algorithm. A hybrid variant seeds the evolutionary population with an A\* route. The benchmark contains 12 generated scenarios spanning three grid sizes, two obstacle densities, and unit-cost or weighted terrain. Across the executed trials, classical methods reached every generated goal. On weighted maps, UCS and A\* recorded the lowest mean path cost among the classical methods. Across 120 evolutionary trials, random initialization succeeded in 73.3% of trials, while A\*-seeded initialization succeeded in 100%. The findings support a narrow conclusion for this benchmark: search-derived initialization improved evolutionary reliability and final fitness, while classical algorithms retained clearer optimality guarantees.

## Keywords

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Graph search; A\*; Dijkstra; Genetic Algorithm; pathfinding; evolutionary optimization; reproducibility

## Project provenance

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The repository originally contained two coursework notebooks. HW01 was an eight-row housing-price regression using normalization and gradient descent. HW02 was an MNIST PyTorch MLP. Because neither notebook implemented graph search, the final project preserves them as legacy references and adds a new, self-contained pathfinding study rather than making an unsupported claim about their contents.

# 1. Introduction and problem statement

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Path planning exposes a useful distinction between finding any feasible route and finding a low-cost route efficiently. Breadth-first and depth-first procedures differ in traversal order. Uniform-cost search accounts for edge cost, while heuristic methods use problem structure to reduce search effort. Evolutionary optimization provides a different mechanism: it explores a population of candidate action sequences and can express penalties beyond a single shortest-path objective.

This study asks whether a classical route can provide a useful prior for an evolutionary optimizer without hiding the cost of obtaining that prior. The experimental object is a four-neighbor obstacle grid. Every method receives the same start, goal, obstacles, and terrain costs. The comparison records route quality and the amount of search effort used to obtain it.

## 2. Research questions and contributions

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**RQ1.** How do uninformed, cost-based, and heuristic methods differ on unit-cost and weighted grids?

**RQ2.** Does A\* seeding change Genetic Algorithm success rate, fitness, convergence, or variability?

**RQ3.** How do grid size and obstacle density affect the observed trade-offs?

The project contributes an executable notebook, a deterministic benchmark generator, comparable metrics for six classical methods, a random-initialized GA baseline, an A\*-seeded hybrid ablation, machine-readable results, and an academic report and presentation generated from the same run.

## 3. Related work

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Dijkstra's shortest-path procedure established a cost-ordered approach for graphs with nonnegative edge costs [1]. Hart, Nilsson, and Raphael formalized heuristic search and the conditions under which A\* can retain an optimality property [2]. Holland's work provides the evolutionary basis for selection, crossover, and mutation used by the Genetic Algorithm [3]. The present study uses these established ideas as baselines and evaluates a deliberately narrow hybrid: an A\* route becomes an informed member of the initial evolutionary population.

## 4. System overview

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Figure 1. Representative obstacle grid. Dark cells are obstacles; the red line illustrates a feasible route between the green start and red goal. The executed notebook generates the benchmark and records each scenario seed.

The workflow is: generate a reachable grid, run all deterministic baselines, run five trials for each GA initialization, write raw results, then generate figures and summary statistics from those files.

## 5. Benchmark and methodology

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The generator creates square four-neighbor grids at 15, 25, and 35 cells per side. It samples obstacle layouts at densities 0.15 and 0.25, reserves the start and goal, and retries with a deterministic seed offset until a route exists. Unit-cost scenarios assign cost 1 to every move. Weighted scenarios assign traversable cells one of the costs 1, 3, or 7. The same grid layout is used to compare the two cost modes.

| Design element               | Configuration                                 |
|------------------------------|-----------------------------------------------|
| Scenarios                    | 12 maps: 3 sizes x 2 densities x 2 cost modes |
| Map seeds                    | 7000 through 7011                             |
| GA seeds                     | 11, 22, 33, 44, 55                            |
| Population / generations     | 30 / 50                                       |
| Elite / crossover / mutation | 2 / 0.80 / 0.05                               |
| Action budget                | 6 x grid size                                 |

## 6. Metrics and fairness controls

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For classical search, the notebook records success, path length, path cost, nodes expanded, nodes generated, maximum frontier size, and runtime. UCS and A\* provide the cost reference on weighted maps because the Manhattan heuristic is admissible when multiplied by the minimum terrain cost. BFS is interpreted as length-optimal only in unit-cost settings.

The GA uses a fixed-length action chromosome. Invalid moves and revisits receive penalties; incomplete chromosomes receive a remaining-distance penalty. Both GA variants use the same population size, generation count, selection, crossover, mutation rate, and five seeds. The hybrid records the A\* seed-generation time separately so the initialization trade-off remains visible.

## 7. Algorithm design

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| Family           | Methods         | Primary comparison                    |
|------------------|-----------------|---------------------------------------|
| Uninformed       | BFS, DFS, IDS   | Feasibility, length, expansion effort |
| Cost / heuristic | UCS, Greedy, A* | Weighted cost and search effort       |
| Evolutionary     | Random GA       | Stochastic route optimization         |
| Hybrid           | A*-seeded GA    | Effect of informed initialization     |

## 8. Classical search results

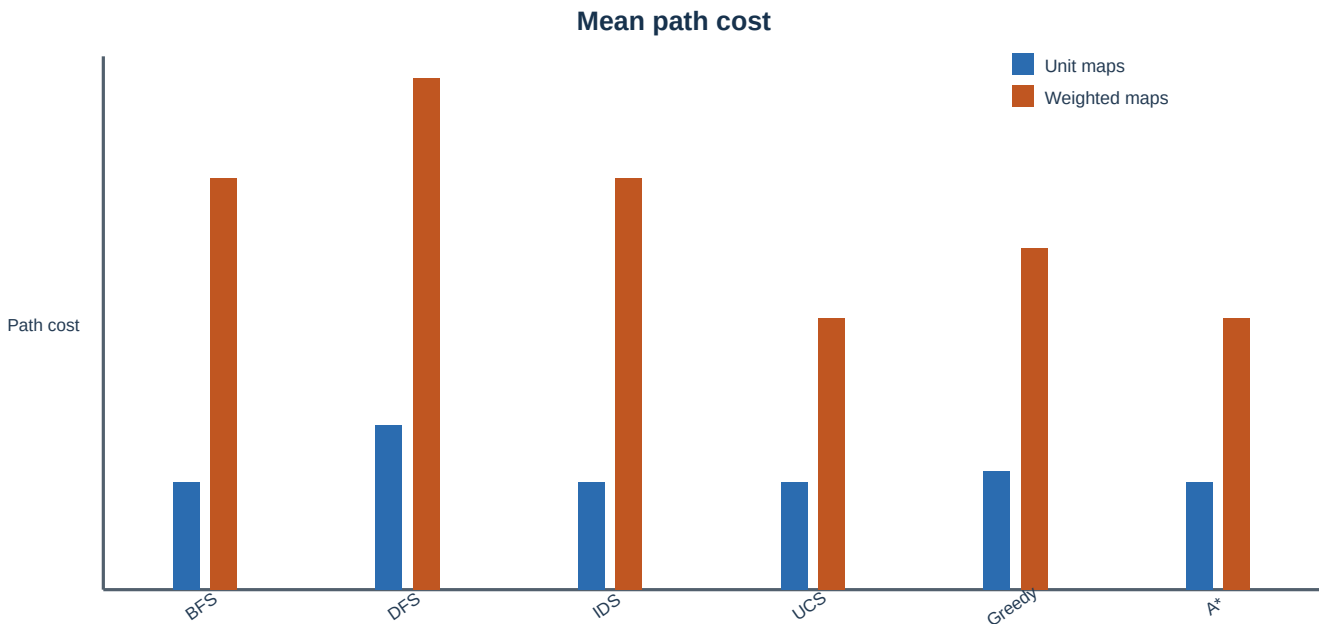


Figure 2. Mean path cost across the 12-map classical comparison. Weighted terrain separates methods that only minimize steps from methods that account for traversal cost.

| Method | Unit success | Unit cost | Weighted success | Weighted cost | Unit expansions |
|--------|--------------|-----------|------------------|---------------|-----------------|
| BFS    | 100%         | 48.33     | 100%             | 184.67        | 552.5           |
| DFS    | 100%         | 73.67     | 100%             | 229.67        | 96.2            |
| IDS    | 100%         | 48.33     | 100%             | 184.67        | 114936.0        |
| UCS    | 100%         | 48.33     | 100%             | 122.00        | 552.5           |
| Greedy | 100%         | 53.33     | 100%             | 153.33        | 68.2            |
| A*     | 100%         | 48.33     | 100%             | 122.00        | 382.7           |

All classical methods found a route on every generated map. On unit-cost maps, BFS, IDS, UCS, and A\* shared the lowest mean path cost of 48.33. On weighted maps, UCS and A\* shared the lowest mean cost of 122.00. DFS and Greedy expanded fewer nodes on average in this benchmark, but their routes were more expensive. IDS found the same unit-cost route length as BFS while expanding substantially more nodes because it repeated bounded-depth searches.

## 9. Evolutionary and hybrid results

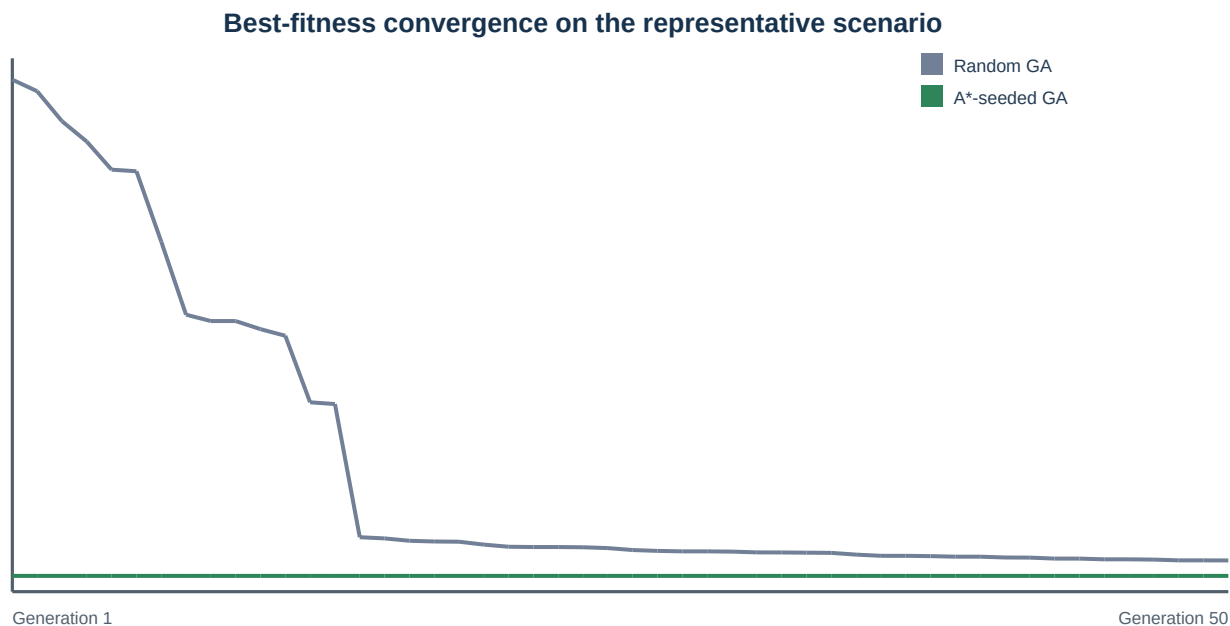


Figure 3. Mean best-fitness trajectory across five seeds on the 25x25, 15% unit-cost scenario. Lower fitness is better.

| Initialization | All success | Mean final fitness | Std. dev. | Mean successful cost |
|----------------|-------------|--------------------|-----------|----------------------|
| Random GA      | 73.3%       | 565.746            | 593.212   | 174.14               |
| A*-seeded GA   | 100.0%      | 87.658             | 50.080    | 85.17                |

Across all 120 evolutionary trials, random initialization succeeded in 73.3% of trials and A\*-seeded initialization succeeded in 100%. Mean final fitness decreased from 565.746 to 87.658. The seeded variant also matched the classical reference cost on successful unit and weighted trials in the aggregate table. These results support the hybrid hypothesis for this benchmark, while the comparison remains conditional on the chosen grid generator, action encoding, and A\* prior.

## 10. Discussion

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The classical results reflect the objectives built into each algorithm. BFS and IDS are aligned with step count on unit-cost grids, but they do not account for weighted terrain. UCS and A\* optimize the weighted objective, and A\* reduces the amount of expansion relative to UCS on unit-cost maps in the aggregate results. Greedy Best-First Search uses the same geometric signal without the accumulated-cost guarantee, which explains its lower expansion count alongside higher path cost.

The GA comparison exposes a different trade-off. Random chromosomes must discover feasible action sequences through mutation and selection, so some trials remain incomplete after the fixed budget. A\* seeding places a feasible, low-cost route in the initial population and preserves it through elitism. This directly improves reliability in the measured benchmark, but it also means the hybrid inherits information from a classical solver. The result should be interpreted as evidence for search-guided initialization, not as evidence that an uninformed GA dominates exact search.

## 11. Limitations and threats to validity

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- The benchmark is synthetic and uses small-to-medium grids, so it does not represent all graph topologies or real routing environments.
- The GA uses a simple fixed-length action chromosome. A different encoding, repair policy, population size, or fitness weighting could change the result.
- Runtime measurements are environment-dependent and should be compared within the same execution environment.
- The seeded method receives an A\* route by design. Its benefit is therefore a hybridization result, not an independent comparison against a purely evolutionary method.
- The experiment reports aggregate means and standard deviations but does not claim statistical generalization beyond the generated scenarios.

## 12. Future work

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Future extensions could evaluate irregular graphs, multi-objective route quality, larger repeated benchmark suites, alternative evolutionary encodings, and a cost-aware repair operator. A stronger study would also pre-register scenario seeds and compare confidence intervals across independently generated map families.

## 13. Reproducibility and repository artifacts

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The executed notebook is the single project implementation. It generates the graph instances, runs every algorithm, writes the result files, and creates the SVG evidence. The repository records the exact scenario configuration in `experiment_manifest.json`. No restricted course dataset is required.

| Artifact                                                      | Purpose                                          |
|---------------------------------------------------------------|--------------------------------------------------|
| <code>src/Graph_Search_Evolutionary_Optimization.ipynb</code> | Complete executed implementation and narrative   |
| <code>results/algorithm_comparison.csv</code>                 | 72 deterministic classical-search rows           |
| <code>results/stochastic_trials.csv</code>                    | 120 GA trial rows with seeds and fitness         |
| <code>results/convergence.csv</code>                          | 6,000 generation-level convergence rows          |
| <code>results/figures/</code>                                 | Generated path, comparison, and convergence SVGs |
| <code>reports/...Report.pdf</code>                            | Academic project report                          |
| <code>reports/...Presentation.pptx</code>                     | Academic presentation                            |

## 14. Conclusion

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This study converted a repository of disconnected coursework artifacts into a reproducible search-and-optimization project. Classical search methods provided dependable reference solutions, with UCS and A\* producing the lowest weighted path cost in the executed benchmark. The A\*-seeded Genetic Algorithm improved evolutionary reliability from 73.3% to 100% successful trials and reduced mean final fitness from 565.746 to 87.658. The conclusion is intentionally bounded: informed initialization helped on these deterministic grids, while exact classical methods remained the clearest reference for optimal path cost.

## References

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- [1] E. W. Dijkstra, "A note on two problems in connexion with graphs," *Numerische Mathematik*, vol. 1, pp. 269–271, 1959. doi: 10.1007/BF01386390.
- [2] P. E. Hart, N. J. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE Transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968. doi: 10.1109/TSSC.1968.300136.
- [3] J. H. Holland, *Adaptation in Natural and Artificial Systems*, University of Michigan Press, 1975. doi: 10.7551/MITPRESS/1090.001.0001.